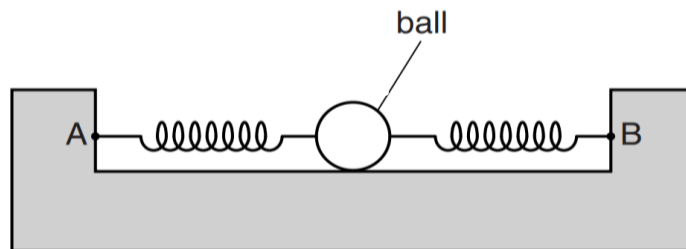


- 3 (a)** A ball is held between two fixed points A and B by means of two stretched springs as shown in Fig. 3.1.



**Fig. 3.1**

The ball is free to oscillate horizontally on the smooth horizontal table. The variation of the acceleration  $a$  of the ball with its displacement  $x$  from its equilibrium position is shown in Fig. 3.2.

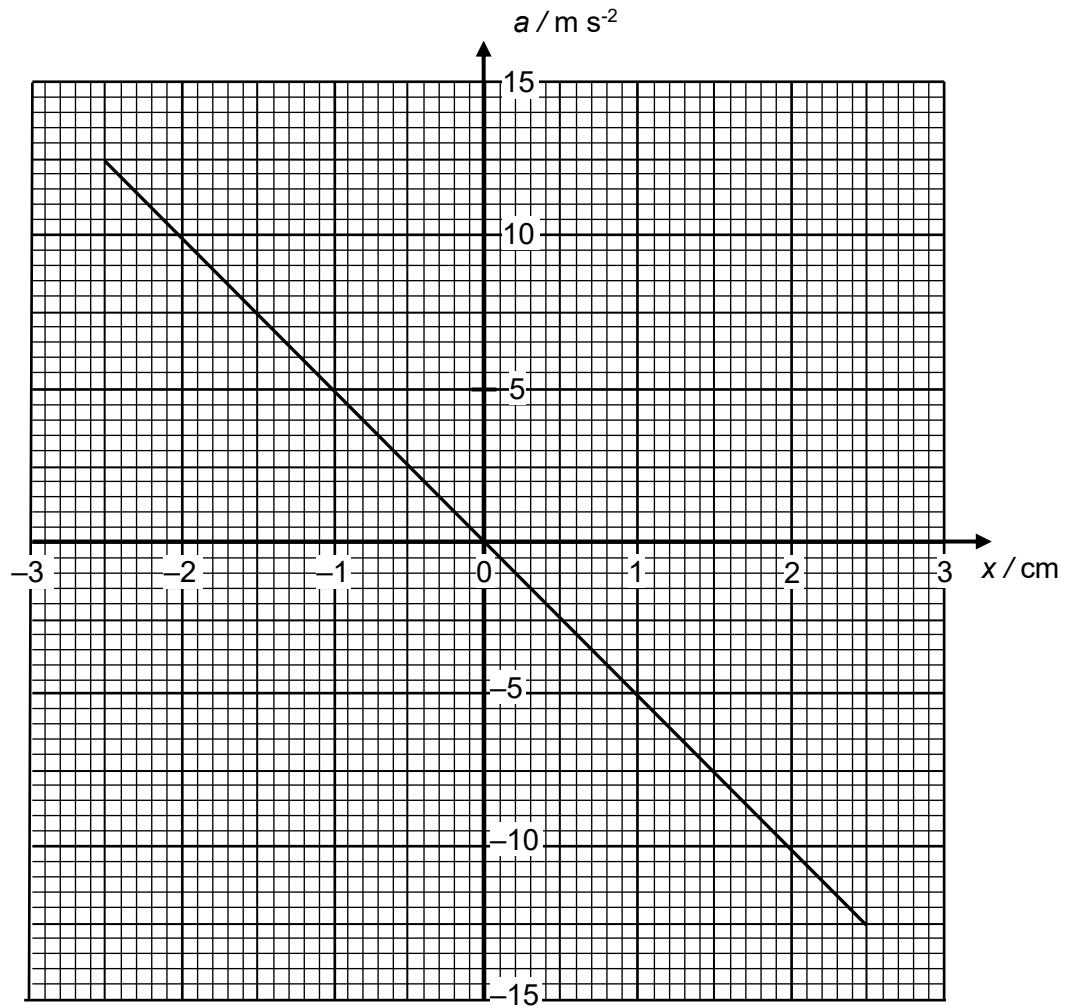


Fig. 3.2

- (i) State and explain the features of Fig. 3.2 which indicate that the moving ball is exhibiting simple harmonic motion.

.....

.....

.....

..... [2]

**Solution:**

The graph is a straight line that passes through the origin, this indicates that the acceleration is directly proportional to the displacement from the equilibrium position. **1**

The gradient of the graph is negative, which indicate that the acceleration and the displacement are always in opposite directions (or the acceleration is always directed to a fixed equilibrium position). **1**

- (ii) On Fig. 3.3, sketch the variation of the velocity  $v$  of the ball with  $x$ . Include the values of the horizontal and vertical intercepts of the graph.

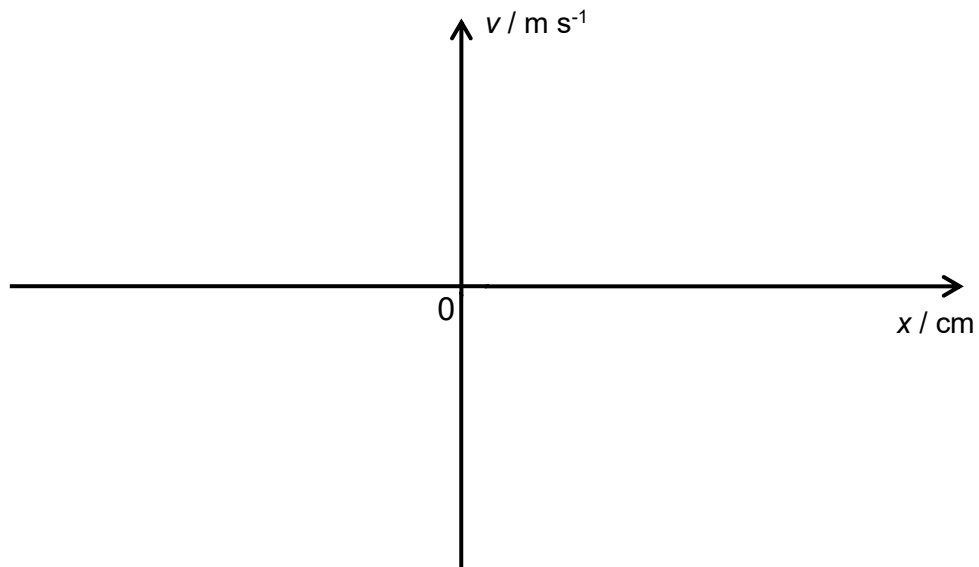
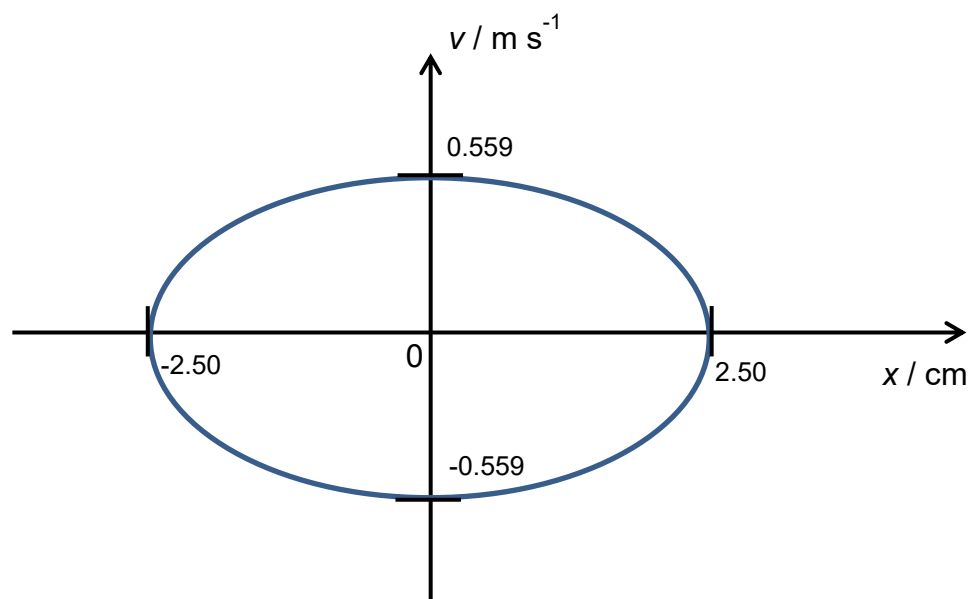


Fig. 3.3

[3]

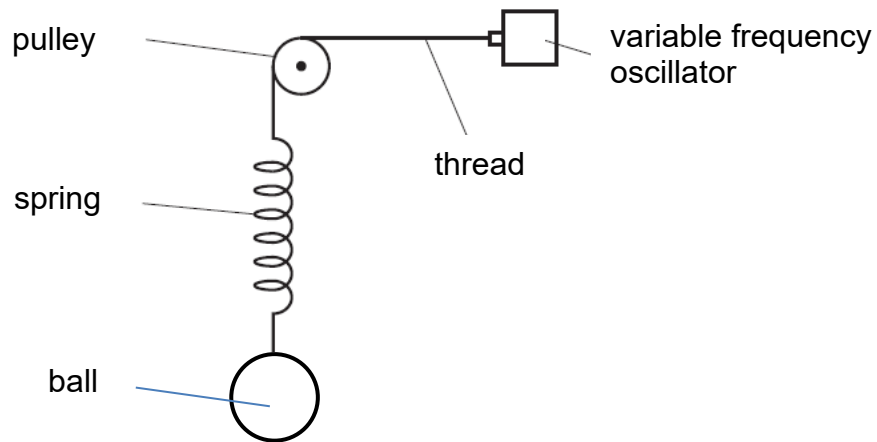
**Solution:**



**Mark scheme:**

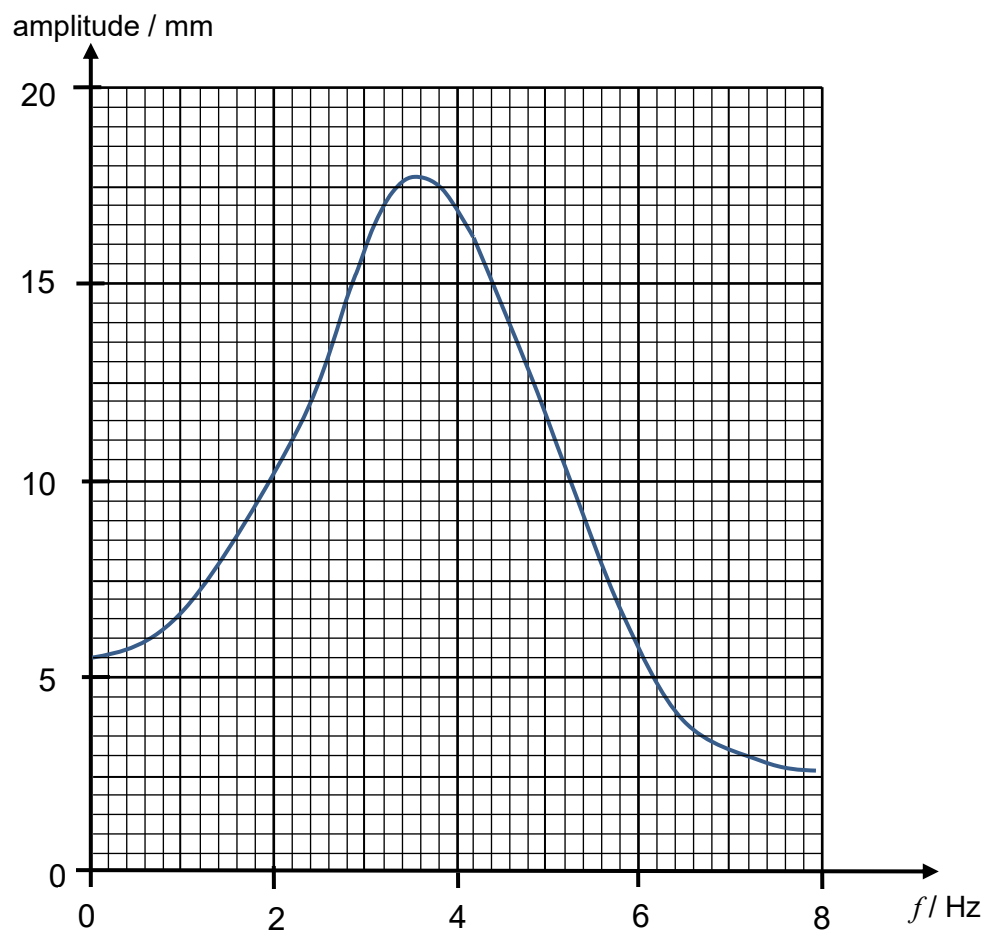
- Shape of graph [ellipse] 1
- From Fig. 3.2,  $x_o = 2.50 \text{ cm} = 0.0250 \text{ m}$   
 ➔ Labelled horizontal intercepts at  $\pm 2.50 \text{ cm}$  1
- From Fig. 3.2,  $a_o = 12.5 \text{ m s}^{-2}$ ,  $a_o = \omega^2 x_o \rightarrow \omega^2 = 12.5 / 0.0250 = 500 \text{ s}^{-2}$ ,
- $v_o = x_o \omega = 0.559 \text{ m s}^{-1}$   
 ➔ Labelled vertical intercepts at  $\pm 0.559 \text{ m s}^{-1}$  1

- (b) The ball is attached to the apparatus illustrated in Fig. 3.4 in order to investigate its vertical oscillations on a spring. The amplitude of the vibrations produced by the oscillator is constant.



**Fig. 3.4**

The variation of the amplitude of the oscillations of the ball with the frequency  $f$  of the oscillator is shown in Fig. 3.5. The oscillations are assumed to be simple harmonic.



**Fig. 3.5**

|  |       |                                                                                                                                                                                                        |     |
|--|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
|  | (i)   | State the natural frequency of the oscillations.                                                                                                                                                       |     |
|  |       | natural frequency = ..... Hz                                                                                                                                                                           | [1] |
|  |       | <b>Solution:</b><br>3.5 Hz                                                                                                                                                                             | 1   |
|  | (ii)  | Explain why Fig. 3.5 shows that the oscillations are damped.                                                                                                                                           |     |
|  |       | .....<br>.....<br>.....                                                                                                                                                                                |     |
|  |       | .....                                                                                                                                                                                                  | [1] |
|  |       | <b>Solution:</b><br>The graph of Fig. 3.5 shows that the resonance occurs at 3.5 Hz with a finite amplitude showing the system is damped. (Without damping the resonance amplitude would be infinite.) | 1   |
|  | (iii) | On Fig. 3.5, sketch a possible variation of the amplitude of the oscillations of the ball with $f$ if the oscillations are damped with a greater resistive force.                                      | [3] |
|  |       | <p>amplitude / mm</p> <p>Fig. 3.5</p>                                                                                                                                                                  |     |

|  |  |                                                                                                                                                                                                                                                                                                               |             |
|--|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
|  |  | <ul style="list-style-type: none"> <li>• Lower resonance frequency <math>\rightarrow</math> 'peak' shifted to lower <math>f</math></li> <li>• Lower amplitude at ALL frequencies (except at <math>f = 0</math>)</li> <li>• Amp unchanged at <math>f = 0 \rightarrow</math> same vertical-intercept</li> </ul> | 1<br>1<br>1 |
|--|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|

|   |     |                                                                                           |
|---|-----|-------------------------------------------------------------------------------------------|
| 4 | (a) | A Keck telescope at Mauna Kea, Hawaii, is the world's largest optical telescope and has a |
|---|-----|-------------------------------------------------------------------------------------------|